Joshua Hall

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He/Him/His

SKILLS

Programming
Java, JavaScript, HTML5/CSS, Python, C, SQLite

Tools & Technologies
VSCode, Bash, Git, Figma, ReactJS, React Native, Expo, Docker

EDUCATION

Bachelor of Science in Physics and Computer Science, minor in Gender and Women's Studies
University of Wisconsin-Madison
08/2021 – 08/2025
Madison, United States

AWARDS

Student Initiative & Innovation Award
Cody Gray - Building Services Manager: Sellery Residence Hall
12/2021

COURSES

Building User Interfaces, Database Management Systems, Intro to Human Computer Interaction, Intro to Artificial Intelligence, Intro to Algorithms, Intro to Computer Engineering, Intro to Discrete Mathematics, Machine Organization and Programming, Software Engineering, Intro to Object-Oriented Programming

PROFESSIONAL EXPERIENCE

University of Wisconsin-Madison
Building Services Student Supervisor
08/2022 – 08/2025

- Trained and supervised a team of ~12 student employees each semester and led team meetings on how to maintain a sanitary residence hall environment.
- Managed building services projects, coordinating staff to prepare facilities for residents on tight timelines.
- Led teams through time-sensitive tasks, making critical decisions during limited staffing or emergencies.
- Collaborated with fellow supervisors and full-time staff, including those with English as a second language, to identify and resolve challenges.

Labor Hour Calculator

- Identified inefficiency in staff hour tracking, proposed a solution to management, and developed an Apps Script automation that integrated seamlessly with existing Google Sheets, simplifying work without disrupting existing workflows.
- Designed logic to handle complex scheduling cases (multi-shift days, time formatting, validation, and automatic lunch deductions) so calculations were accurate and supervisors didn't need to fix errors manually.
- Used regularly by my manager and supervisors, eliminating manual tallying and saving 4 hours per month while reducing errors.

University of Wisconsin-Madison
Building Services Student Custodian
08/2021 – 05/2022

- Maintained a safe and sanitary environment in University residence halls.
- Received the Student Initiative & Innovation Award for strong work ethic in winter of 2021.

PROJECTS

Badger Chatroom

- Designed and implemented the frontend for a chatroom web application that recreates the feel of an old-school forum for UW–Madison students.
- Built with ReactJS using component-based architecture, client-side routing (BrowserRouter), and state management to load chatroom messages dynamically.
- Integrated a REST API to fetch/post only the active room's current page, minimizing network and render work.
- Ensured a smooth user experience with reload-free room switching, simple message posting via useRef-based inputs, consistent styling with Bootstrap, and clear validation/error feedback for login and registration.

Minimal Heap Allocator

- Developed a custom heap allocator in C to manage dynamic memory, using a best-fit strategy with coalescing to reduce fragmentation.
- Designed allocation logic to split or merge blocks efficiently, ensuring minimal wasted space and predictable performance.
- Validated reliability by handling edge cases (exact fits, split vs. absorb, end of memory) and confirming correct behavior across many allocation scenarios.

Custom Hash Table

- Implemented a custom hash table in Java with separate chaining and automatic resizing to maintain fast lookups and inserts as data grows.
- Designed as a building block for fast key–value access (e.g., sessionId to user session), used in caches, session stores, and in-memory indexes.
- Ensured reliability under collisions and invalid inputs.
- Validated functionality through targeted JUnit tests, confirming correct behavior for insertion, deletion, resizing, and collision handling.